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Public Speakers With Nonnative Accents Garner Less Engagement

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Abstract

Can nonnative English accents become barriers to garnering attention in public discourse? The current study examined this question. Analyzing 5,367 TED Talks through computational methodologies such as voice recognition, natural language processing, and vision models, we investigated the relationship between speakers' accents and online engagement. After adjusting for various control variables with a series of robustness checks, we found a sizeable disparity in public discourse: Speakers with nonnative accents received less engagement than speakers with native accents. To complement our findings, we conducted a controlled social-psychological experiment among English-speaking American adults ($N = 462$) and a direct replication ($N = 916$) that corroborated our computational analyses and highlighted stereotyping and processing disfluency as key factors driving reduced engagement in accented speakers. Our research highlights the pervasive impact of accent discrimination in global communication and emphasizes the need for strategies to mitigate its detrimental effects on knowledge exchange across cultural and linguistic boundaries.

Keywords

accent, computational methods, text analysis, stereotypes, public discourse

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Today, with the rise of globalization, people are more connected than ever: They can discuss new ideas, convince political rivals, tune in to public discourse, and learn about creative ideas. Increasing globalization and ease of access to the Internet have facilitated the participation of people from diverse backgrounds in public discourse. Public speaking—whether to inform, entertain, or motivate—has facilitated a fast-paced dissemination of novel ideas, groundbreaking research, creative arts, and insightful stories. Public speakers compress their years of work into short speeches that can garner millions of views in a matter of days or even hours. For example, TED Talks—videos from expert speakers on topics as diverse as education, business, science, technology, and creativity—are viewed by millions of people around the globe every day (Anderson, 2017). TED may be one of the most successful online producers of scientific and technical videos (Sugimoto et al., 2013).

Public speakers, such as TED speakers, are known for their engaging, inspiring, and informative content; however, they do not operate in a social vacuum. They bring their cultural background to the social arena, and their very presence could evoke particular emotional reactions. One highly conspicuous verbal cue to people's cultural background is their spoken accent. Individuals can immediately pick up whether someone is a "native" (or standard-accented)[1] speaker or not, often outside of their conscious awareness (Acheme & Cionea, 2022). This might be the case because humans begin to show an affinity for particular kinds of language from an early age, with infants gravitating toward individuals who speak their mother tongue (Kinzler et al., 2007; Kuhl et al., 2003).

In public discourse, an individual's language (including their dialect, accent, or pronunciation) conveys a

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large amount of social cues to the listener (Kinzler, 2021; Ladegaard, 1998). People might quickly pick up accent-related cues to categorize people into "friends" or "foes," hence stereotyping the speaker's intentions, goals, and identity as well as their intelligence, competence, and expertise (Fuertes et al., 2012; Kinzler et al., 2007; Lippi-Green, 2011).

Unfortunately, language and accent are "often neglected in the vast literature on person perception and social categorization" (Kinzler, 2021, p. 242). This may be due to scholars prioritizing other visible or more overt characteristics (e.g., physical appearance) and other forms of nonverbal features (e.g., gender), inadvertently overlooking the subtleties of language and accent in impression formation and engagement. This pattern of neglect can be found in surveys of publications on language over the last 3 decades in which accent was notably underexamined (Gasiorek et al., 2012). Previous work has uncovered a variety of biases associated with accents, a kind of bias sometimes referred to as "accentism" (Hegarty, 2020; Matsuda, 1991). Importantly, although accent-based bias may be correlated with other kinds of bias on the basis of social-group membership (e.g., class, race, gender), it is a distinct form of discrimination (Hegarty, 2020).

There are two viable social-psychological explanations to explain the link between nonnative accents and nondesirable outcomes (Foucart et al., 2020; Spence et al., 2022). The first explanation is a "processing fluency" hypothesis, according to which people with nonnative accents meet nondesirable social outcomes because nonnative-accented speech is cognitively taxing to process (Dragojevic & Giles, 2016; Floccia et al., 2009; Hideg et al., 2022). After all, easier-to-process stimuli are perceived to be more truthful (Reber & Schwarz, 1999) and credible (Lev-Ari & Keysar, 2010) and elicit more positive evaluations (Dragojevic, 2020) than more cognitively taxing stimuli. Thus, how much cognitive effort an audience expends to process content provided by a speaker may impact their engagement. The other explanation is based on stereotyping. This explanation is predicated on the notion that accents tie people to their social identities. The idea is that nonnative accents signal "otherness" and increase stigma (Gluszek & Dovidio, 2010; Spence et al., 2022), which might lead to stereotyping the speaker. Such social stereotypes can be captured along the dimensions of warmth and competence (Cuddy et al., 2008). Because stereotyping shapes how a speaker is perceived, this bias may act as a filter, shifting attention away from the content and toward characteristics of the speaker.

Many public speakers around the globe come from higher socioeconomic strata, have technical command of English, possess some level of public fame, and speak with substantially different accents. Nonnative accents (e.g., Indonesian, Nigerian, Mexican) can particularly penalize the speaker in the public arena. Many people who speak English were not raised in English-speaking countries (e.g., the United States, Australia, England) but are considered native speakers. It remains unknown to what degree and why public speakers with nonnative accents might be penalized (in terms of attracting engagement on the Internet). Here, we quantified how much less engagement (i.e., views and likes) public speakers with nonnative accents pull compared with speakers with native accents and how we can explain this reduced engagement from a social-psychological perspective. In Study 1, we used a variety of computational technologies adapted from image processing, voice recognition, and natural language processing (see Fig. 1) to test our hypothesis that, across thousands of TED Talks, speakers with nonnative accents would garner less online engagement. Complementing our computational analyses, Study 2 involved a social-psychological experiment to understand and compare our proposed mechanisms (i.e., the processing fluency and stereotyping accounts) at play to explain the accentism bias.

Research Transparency Statement

General disclosures

Conflicts of interest: All authors declare no conflicts of interest. **Funding:** This research received no specific funding. **Artificial intelligence:** No AI-assisted technologies were used to conduct this research or to create this article. **Ethics:** This research received approval from the ESSEC Business School ethics board (ID reference: 2024-10). **Open Science Framework (OSF):** To ensure long-term preservation, all OSF files were registered at https://osf.io/kt85s. **Computational reproducibility:** The computational reproducibility of the results in the main article (but not in the Supplemental Material available online) has been independently confirmed by the journal's STAR team.

Study 1 disclosures

Preregistration: No aspects (hypotheses, method, and analysis plan) of the study were preregistered. **Materials:** The materials used in Study 1 are publicly available (https://osf.io/thqfw). **Data:** All primary data are publicly available (https://osf.io/4uvmf). **Analysis scripts:** All analysis scripts are publicly available (https://osf.io/e2awr).

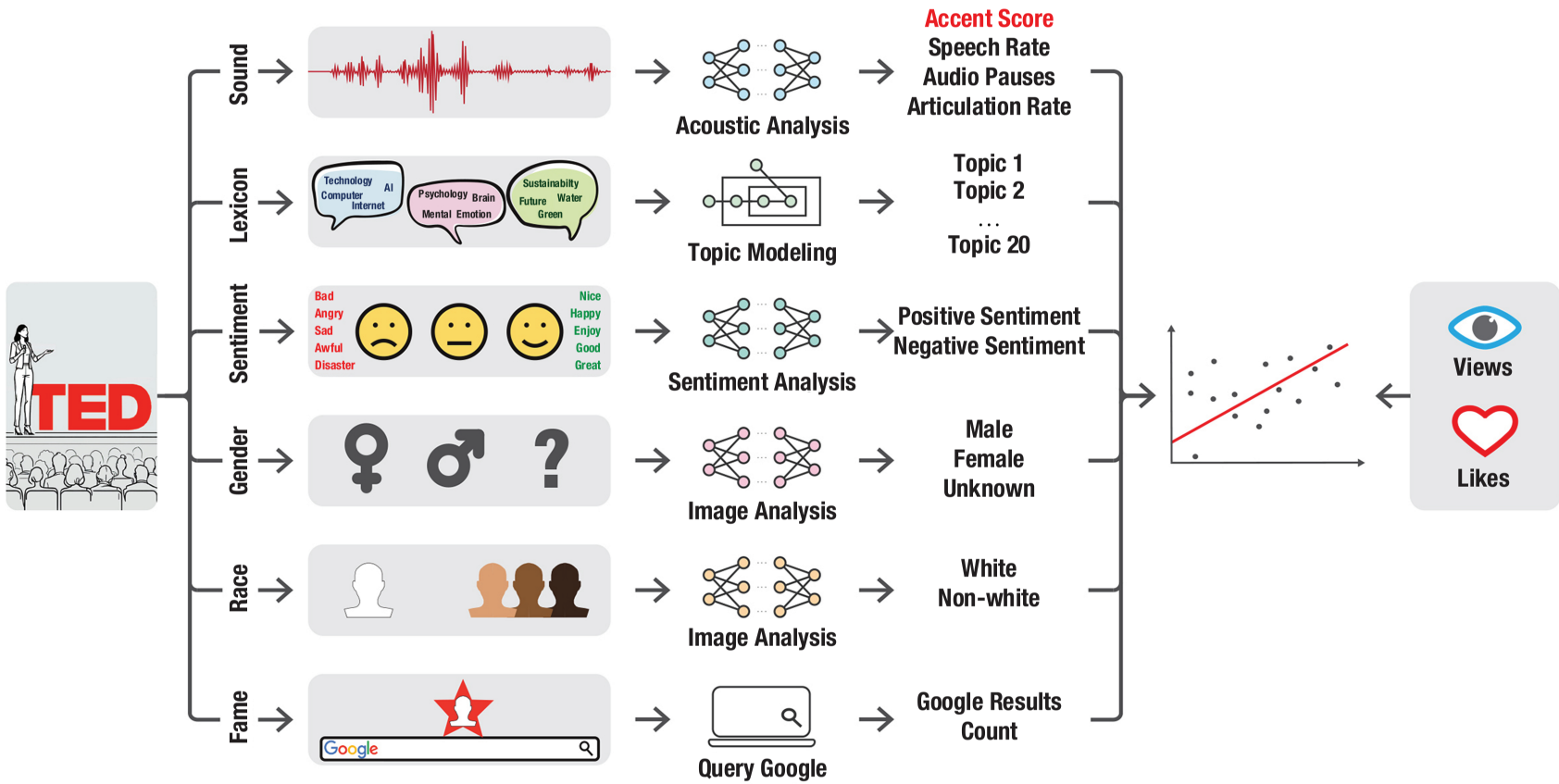

Fig. 1. Schematic representation of the multimodal analyses used to assess whether nonnative accents garner less online engagement. The main independent variable (accent score) is shown in red. To account for potential confounds, we quantified and controlled for other acoustic features of the voice, the topics discussed, the emotional tone of the talks, and each speaker's gender, race, and fame. Note that the regression line is simply an icon of linear models; it is not based on our data.

Study 2a disclosures

Preregistration: No aspects of this study were preregistered. Data collection began on January 23, 2024. **Materials:** All study materials are publicly available (https://osf.io/yrvgq). **Data:** All primary data are publicly available (https://osf.io/5s2vu). **Analysis scripts:** All analysis scripts are publicly available (https://osf.io/73t2u).

Study 2b disclosures

Preregistration: The research aims/hypotheses, method, and analysis plan were preregistered (https://osf.io/zmr2n) on July 24, 2024, prior to data collection, which also began on July 24, 2024. There were no deviations from the preregistration. **Materials:** All study materials are publicly available (https://osf.io/t25j8). **Data:** All primary data are publicly available (https://osf.io/arfks). **Analysis scripts:** All analysis scripts are publicly available (https://osf.io/mgd8y).

Study 1: Accentism on the Stage

Method

TED Talks corpus. We obtained information on 5,367 TED Talks from TED.com, a leading platform dedicated to disseminating novel ideas around the globe. The context of TED Talks was particularly suitable for our analysis because of the reach of the platform coupled with the variety of information available (see Fig. 1). In addition, as outlined in Table 2 in the Supplemental Material, viewership is dominated by young Americans. TED.com provides information on the textual content of each talk (used for text analysis), the image of the speaker (used to determine the speaker's race), name of the speaker (used to classify gender), and the audio of the talk (used to extract accent and other acoustic features). We also recorded the length of each talk, allowing us to control for the talks' duration, and the amount of time each talk has spent on TED.com, which might affect the number of views or likes it gets. We controlled for the fixed effect of publication year because the platform might have had different marketing strategies and budgets each year (e.g., more or less advertising on social media). The fame of each speaker was also used to account for how a speaker's popularity affected levels of online engagement. Furthermore, we conducted a sentiment analysis on each TED Talk to ensure that reduced engagement was not primarily due to the valency of the speaker's content. Similarly, we also used topic modeling to control for the topic of the speaker's talk. Our entire corpus of 5,367 TED Talks had 9,272,328 words (M = 1,727.66) overall.[2]

We had two dependent variables of interest. The first was the number of views each talk received (M = 2,178,770, Md = 1,421,580). This number was skewed, so we log-transformed it ($\text{skewness}_{\text{pre}}$ = 8.14, $\text{skewness}_{\text{post}}$ = −1.54). Our second dependent variable was the number of likes each talk garnered (M = 64,832, Md = 42,000). The number of likes was also skewed, so we log-transformed it ($\text{skewness}_{\text{pre}}$ = 7.95, $\text{skewness}_{\text{post}}$ = −1.54).

Automated accent detection. We obtained pronunciation scores for each speaker using the MyProsody package (Version 10; Kane et al., 2024) in Python. This package uses automatic speech recognition and the hidden Markov model to train a model for each phoneme, considering features such as phoneme mispronunciation and prosodic stress to score speakers' pronunciation scores. Each speaker received a pronunciation score between 0 and 100. We reverse-coded these scores by subtracting each score from 100 to get a measure of the nonnative accent score (i.e., pronunciation scores of 0 and 100 indicating a high native accent and high nonnative accent, respectively; for a distribution of the accent scores, see Fig. S2 in the Supplemental Material). We validated the scores provided by this package on the George Mason University speech accent archive data set (Weinberger, 2015) with 643 native and 1,103 nonnative audio files as ground truth, $t(1743) = -12.61$, $p < .001$, $M_{\text{native}} = 79.05$, $M_{\text{nonnative}} = 88.41$. For illustration, a nonnative speaker talking about dishonesty in children (Lee, 2016) received an accent score of 30.29, and a native educator talking about the riddle of multiplying rabbits (Gendler, 2019) received an accent score of 0 (M = 12.23, Md = 5.15). Other acoustic and vocal features of the speaker might influence how much a talk is liked and viewed. To isolate the effect of these variables in our regression framework, we controlled for pauses, speech rate (the number of syllables divided by the total audio duration), and articulation rate (the number of syllables divided by the active phonetic duration).

Topic modeling. To control for the topics discussed in each talk, we used latent Dirichlet allocation (LDA), a common topic-modeling technique (Blei, 2012). We used LDA implementation from Python's sklearn package (Version 0.24.2; Pedregosa et al., 2011). The topics were estimated with an online variational Bayes algorithm (M. Hoffman et al., 2010; M. D. Hoffman et al., 2013). We then performed LDA with eight to 50 topics and then selected 20 on the basis of optimal log-likelihood scores. Log likelihood measures how plausible model parameters are given the data, and a larger log-likelihood score shows a better fit. Examples of topics discussed and their

sample comprising words are reported in Figure 1 in the Supplemental Material.

Sentiment analysis. Beyond the topics covered in each talk, the overall sentiment of a talk (e.g., sad, happy) may be considered an important covariate. Talks that have a less positive sentiment might receive fewer views, for example, and may be more likely to come from speakers with a nonnative accent (e.g., refugees sharing their stories). To address this possibility, we controlled for the sentiment of each talk using the valence measure provided by the TextBlob package (Version 0.17.1; Loria, 2018) in Python. This algorithm is sensitive to negations (e.g., "not sad") and intensifiers (e.g., "very sad").

Speaker race and gender. To rule out the possibility that accent-based bias could be explained away by racial and gender biases, we included these covariates in our models. We used the deepface package (Version 0.0.75; Serengil & Ozpinar, 2020) in Python to categorize the race of the speaker from their published photo on the TED platform. This package, which wraps several advanced models, has been validated by the University of Massachusetts Amherst Labeled Faces in the Wild data set and has been used by previous researchers to reliably detect race (Huang et al., 2007). Because variations across different non-White racial categories were not part of our work's focus, we dichotomized race into "White" and "non-White" categories. The speakers' genders might also confound potential biases in views and likes, so we extracted from their first names using the gender-guesser package (Version 0.4.0; Michael, 2013) in Python. The package could not identify the gender of 696 names, which we categorized as "unknown." A limitation of this approach is that we had binary labels and some unknown categories based on names. However, the gender classification within the underlying data set of this package has been shown to be of high quality, having undergone manual verification by native speakers from various countries (Santamaría & Mihaljević, 2018).

Speaker fame. The speaker's fame might confound potential accent-driven biases. For example, famous speakers may get more views regardless of their accents. As a proxy for speaker fame, we queried Google's search engine for each speaker and used Google's results count as a measure of their public fame (Sepehri et al., 2025).

Time. We recorded the duration of each video from TED.com, thus ensuring that the length of a speaker's talk was not a driving factor in reducing engagement levels. The publication year and number of days on the platform were documented to confirm that the views and likes a speaker received were not solely determined by how long their talk was published.

Manual annotations. We recruited and trained two undergraduate students whose first language was English to listen to at least 1 min of each TED Talk in our data and annotate the speaker on their accent origin, race, and gender (more than 100 hr of annotation work overall). Annotations were checked by A. Zewail for consistency when there was a lack of clarity or certainty. Before beginning the annotations, we developed a short annotation manual, which is available in the Supplemental Material. Because we used automated methods to classify the speakers' gender and race, which are known to have biases, we used human-annotated categories to benchmark the automatic classifications. The performance of automatic classifiers for gender (F1 score = .93) and race (F1 score = .75) was acceptable. We note the limitation that both of these classifiers output binary results (man and woman for gender and White and non-White for race). Our regression results were robust regardless of whether we incorporated the automatically classified or the human-annotated version of our gender and race variables. We also put a fixed effect of accent origin (11 groups of accents based on our annotation manual) to examine which groups of accents were particularly penalized and which "standard" accents (e.g., American vs. British) were more favored.

Results

After acquiring 5,367 talks from TED.com, we first applied a computational technique to automatically quantify accentedness as well as other acoustic features such as speech rate, audio pauses, and articulation rate. We then conducted an LDA (Blei, 2012) on the text of all talks to quantify the topics that speakers talked about. Using computational image-processing methods, we also classified the binarized gender (male, female, unknown) and race (White, non-White) of the speakers. Because most classifiers insert human biases in their classifications (Davani et al., 2023), we also hand-annotated the entire data set for gender and the kind of accent on the basis of a typology we developed for this study (see the Supplemental Material). We quantified engagement in two different ways: the number of views that a talk has garnered and the number of likes (which, unlike views, require signing into TED.com). Our main independent variable was our computationally derived and validated metric of a nonnative English accent (accent score; see Fig. 1). To rule out other explanations, we also quantified other audio features, lexical content of the talk, sentiment, and fame of the speaker, as described in the Method section.

Table 1. Study 1 OLS Regression Estimates

	Views (log) Estimate (SE)	Likes (log) Estimate (SE)
Accent score	−0.068***	−0.068***
	(0.020)	(0.020)
Pauses	0.0002	0.0002
	(0.0002)	(0.0002)
Speech rate	−0.081*	−0.082*
	(0.034)	(0.034)
Articulation rate	0.018	0.019
	(0.027)	(0.028)
Speaker fame	0.000	0.000
	(0.000)	(0.000)
Sentiment	−0.249	−0.245
	(0.225)	(0.226)
Duration of the talk	0.0001	0.0001
	(0.0001)	(0.0001)
Days on platform	0.0004**	0.0004**
	(0.0001)	(0.0001)
Race (White vs. non-White)	0.072*	0.073*
	(0.031)	(0.031)
Gender (man vs. woman)	0.012	0.012
	(0.027)	(0.027)
Intercept	12.591***	9.059***
	(0.732)	(0.737)
Adjusted R^2	.121	.122

Note: To account for potential confounds, for each TED talk we quantified and controlled for other acoustic features of the speaker's voice, topics discussed, emotional tone, gender, race, fame of the speaker, duration, publication year, and accent groups. These regressions included fixed effects for topics, publication year, and accent group. After controlling for these various factors, we still observed an accent-based penalty in the number of views and likes a talk received. Our results revealed an approximate 7% decrease in view and like count for every 1-*SD* increase in accent score. This decrease translated to losing an average of 93,400 views and 2,800 likes for a single unit increase in the accent score. OLS = ordinary least squares.
*$p < 0.05$. **$p < 0.01$. ***$p < 0.001$.

We ran a series of ordinary least squares regression models to investigate whether speakers' nonnative accent influenced the level of engagement they received. First, we regressed engagement on standardized accent scores and found that speakers with higher accent scores ("stronger" accent) received fewer views ($\beta = -0.18$, $SE = 0.020$, $p < .001$) and fewer likes ($\beta = -0.18$, $SE = 0.020$, $p < .001$) than speakers with lower accent scores. The difference between the effect of accent on views and likes was negligible. Second, we added a fixed effect of accent groups (e.g., Eastern European and Middle Eastern) to account for linguistic differences. After controlling for the fixed effect of the accent group, we found that speakers with higher accent scores still received fewer views ($\beta = -0.13$, $SE = 0.021$, $p < .001$) and likes ($\beta = -0.13$, $SE = 0.021$, $p < .001$). This analysis revealed that, compared with speakers with African accents, speakers with U.S. American (including Canadian—$\beta_{views} = 0.36$, $SE = 0.129$, $p = .005$; $\beta_{likes} = 0.36$, $SE = 0.129$, $p = .005$) and Southern European ($\beta_{views} = 0.63$, $SE = 0.242$, $p = .009$; $\beta_{likes} = 0.64$, $SE = 0.242$, $p = .008$) accents received significantly more likes and views, and speakers with a Southeast Asian accent were the most penalized group ($\beta_{views} = -1.67$, $SE = 0.586$, $p = .005$; $\beta_{likes} = -1.67$, $SE = 0.586$, $p = .004$; see Fig. 2).

We then included a wide array of computationally derived robustness checks to rule out racism, sexism, public fame, speech duration, lexical topics, sentiment (tone), publication date, number of days on TED.com, and other acoustic features of the speaker's voice (i.e., audio pauses, speech rate, and articulation rate). After controlling for these covariates in addition to the manually coded accent groups (see Table 1 and Fig. 2), we found that speakers with higher accent scores received fewer views ($\beta = -0.068$, $SE = 0.020$, $p < .001$) and likes ($\beta = -0.068$, $SE = 0.020$, $p < .001$) than speakers with lower accent scores. These results show that the accent penalty effect was robust to a wide range of control variables, including the strength of the accent present in speakers' speech. In other words, a "stronger" Chinese accent was penalized more than a less conspicuous Chinese accent. Notably, an approximate 7% decrease in either outcome variable translated to losing an average of about 93,400 views and 2,800 likes for every 1-*SD* increase in accent score, highlighting the severity of the engagement penalty for the speaker's accent. In large-scale media environments in which content can receive millions of views, even small proportional changes can translate into large absolute differences in visibility, especially when accumulated over time.

Study 2: Accentism in the Lab

The results from Study 1 were informative and highly ecologically valid. In a sense, we documented accentism "in the wild." However, these computational analyses did not inform us about any potential mechanisms that would account for this accent-based penalty in online engagement. In Studies 2a and 2b, we conducted a social-psychological experiment and direct replication to test the main effect of TED speakers with "nonstandard" (vs. standard) accents receiving less engagement in controlled settings. Furthermore, we aimed to simultaneously test two parallel mechanisms accounting for this effect: the processing fluency and stereotyping accounts. Because specific and well-known accents (e.g., Italian, French, Spanish) are linked to particular stereotypes (Cargile et al., 2010), we aimed to use an accent that was less known to American adults who speak English as their first language. For this reason, we chose the Persian accent, which is noticeably

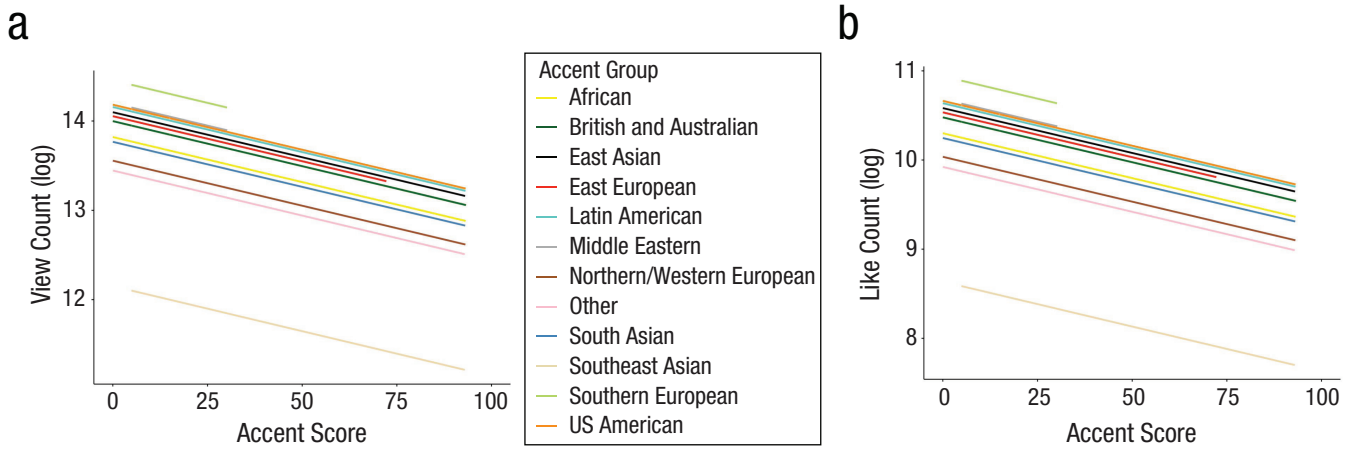

Fig. 2. The relationships between TED Talks speakers' nonnative accent and the level of engagement. After controlling for the fixed effect of the accent group, speakers with higher accent scores still garnered fewer (a) views ($\beta = -0.13$, $SE = 0.021$, $p < .001$) and (b) likes ($\beta = -0.13$, $SE = 0.021$, $p < .001$) than speakers with lower accent scores. Speakers with a Southeast Asian accent were the most penalized group.

recognizable as a nonstandard accent for Americans but is not as widely identifiable.

We selected the text of a short snippet of a TED Talk about creativity and motivation and asked a bilingual, Canadian-Iranian person to record an audio of the snippet as if they were presenting a TED Talk. We ensured that both versions of the talk were of similar length and that they varied only in terms of the accent. Before we conducted the experiment, three bilingual speakers of English and Persian reviewed the stimuli and confirmed that they differed only in terms of the accent and that the Persian accent was natural. We told American English-speaking participants that they were going to listen to a short part of a TED Talk. Participants were randomly assigned to either the American/Canadian accent or the Persian accent (see the Method section). We then asked them how likely they were to show interest, like, or share the talk as a proxy for engagement. We also asked questions about processing fluency and stereotypes associated with the speaker (no visual feature of the speaker was shown to the participants).

Study 2a

Method. We conducted a power analysis using G*Power to detect a small correlation (r) of 0.15; with an alpha level of 0.05 and at 90% power, we needed 463 participants. To ensure we had enough participants after removing those who failed our inclusion criteria, we recruited a total of 492 participants using the Prolific Academic platform. After excluding those who failed to complete the survey (n = 10), those who failed an attention check (n = 11), and those whose first language was not English (n = 9), a total of 462 participants were retained for statistical analyses (270 women, 186 men, five nonbinary, one agender). The average age for our participants was 39.44 years (SD = 12.81). All participants were adults based in the United States. After consenting to the study, participants were asked whether they had a functioning speaker, and on confirmation they were randomly assigned to one of two conditions. In the standard accent condition, the speaker had an American/Canadian accent and presented a brief presentation (audio only) about motivation (length = 39 s). In the nonstandard accent condition, participants were provided with an identical presentation (an audio clip of similar length), but the same speaker spoke with a Persian accent instead. We chose the Persian accent because it is noticeably nonnative but is not widely recognized in the United States (the top 10 languages spoken in U.S. households other than English are Spanish, Chinese, Tagalog, Vietnamese, Arabic, French, Korean, Russian, Haitian, and Portuguese, respectively; Batalova, 2025). To gauge interest, we asked participants to rate how interesting they found this part of the talk on a scale from 1 (*not at all*) to 7 (*very much*). We used the same prompt and scale to gauge how much the participants liked the audio snippet. To measure the participants' interest in sharing the talk, we asked participants to rate how likely they would be to share the talk on social media using the same scale. Because the questions we asked were highly correlated, we averaged them to get a composite measure of "engagement" (α = .90). We repeated all our analyses for each of these three variables separately and report these results in the Supplemental Material.

We measured the ease or difficulty of processing the information in the audio snippet with three questions. All three questions were rated on a scale from 1 (*not at all*) to 7 (*very much*) in a matrix table format. The overall processing fluency score was highly reliable (α = .75). To measure stereotypes, we relied on the

stereotype content model (Fiske et al., 2002). This pluralistic model of stereotyping goes beyond categorizing people on a "goodness-badness" dimension. Rather, the model has two core facets of general stereotype content: warmth (e.g., friendly, good-natured) and competence (e.g., intelligent, skillful; Cuddy et al., 2008; Fiske et al., 2007). This two-dimensional model has been highly generative and powerful in person perception (Fiske et al., 2002, 2007). Participants rated their perception of the speaker's warmth and competence using ratings of two characteristics for each. For warmth, participants rated the speaker on being "warm" and "friendly" (α = .92), and for competence, the speaker was rated on being "knowledgeable" and "competent" (α = .90). Correlations between these variables can be found in the Supplemental Material.

Last, for our manipulation check, we asked participants at the end of the survey to rate how much they agreed that the speaker was a native English speaker on a scale from 1 (*strongly disagree*) to 7 (*strongly agree*). For consistency in accent scores across our studies, we reverse-coded these scores so that a score of 1 would indicate a standard accent and a score of 7 would indicate a nonstandard accent. At the end of the survey, the participants had the opportunity to describe the speaker in their own language in two to three sentences. An exploratory text analysis of their descriptions (detailed in the Supplemental Material) revealed descriptions mainly focused on the actor's voice and accent. This emphasis likely resulted from a lack of visual information and our survey items related to processing fluency rather than the participants' awareness of the experiment's purpose. We ran a multiple mediator model using the psych package (Version 2.4.3; Revelle, 2024) in the R programming language. Our mediation analysis (Fig. 3c) investigated how a speaker's accent (standard vs. nonstandard) influenced the likelihood of engagement on accounts of processing fluency and stereotyping. Bootstrapping was conducted to estimate the standard errors and 95% confidence intervals (CIs) of our mediation effects.

Results. First, we examined our manipulation check to ensure that people could recognize that the accented version was, in fact, not in a standard American/Canadian accent. We asked people to indicate whether they agreed that the speaker was a native English speaker on a scale from 1 (*strongly disagree*) to 7 (*strongly agree*). The standard accent condition (M = 2.66, Md = 2.00) was rated substantially lower than the nonstandard accent condition (M = 5.93, Md = 6.00), t = −22.84, p < .001, Hedge's g = −2.12, 95% CI = [−2.35, −1.89]. Intended engagement was found to be significantly higher in the standard accent condition (M = 3.89) than in the nonstandard accent condition (M = 3.59), t(459.55) = 2.06, p = .040, Hedge's g = 0.19, 95% CI = [0.01, 0.37] (Fig. 3b). Including gender, age, educational background, and political ideology in a regression framework did not change the effect (β = −0.31, SE = 0.145, p = .035).

Participants' intention for engagement was negatively associated with difficulty in cognitive processing (r = −.30, p < .001) and positively associated with perceiving the speaker as competent (r = .66, p < .001) and warm (r = .66, p < .001). Turning our attention to mediating mechanisms (see Fig. 3c), we saw that a talk that was harder to process and given by a speaker perceived as cold and incompetent was least likely to receive engagement. A mediation model with three mediators suggested that the entire mediation model was significant, *ab* effect through all mediators = −0.3, SD = 0.11, 95% CI = [−0.51, −0.09]. Warmth, bootstrapped *ab* effect = −0.15, SD = 0.06, 95% CI = [−0.27, −0.05], and processing fluency, bootstrapped *ab* effect = −0.07, SD = 0.03, 95% CI = [−0.14, −0.01], but not competence, bootstrapped *ab* effect = −0.07, SD = 0.05, 95% CI = [−0.19, 0.03], had significant mediating effects. The mediation model suggested that accent was linked to a combination of cognitive processing and stereotyping (especially perceiving accented people as colder), which were in turn associated with lower engagement.

Study 2b

We directly replicated Study 2a to assess the consistency and reliability of its findings. Overall, we observed a similar pattern in which the nonstandard accent condition received significantly lower intended engagement compared with the standard accent condition. Moreover, results from the mediation analysis reiterated that cognitive processing and stereotyping were two mechanisms that drove the reduced intended engagement within the nonstandard accent condition, highlighting the robustness of our results.

Method. In our preregistration for this study, we conducted a power analysis (with the targeted power being 90%) using the effect size observed in the results of Study 2a (Hedges's g = 0.21). On the basis of these calculations, we aimed to recruit 955 participants on Prolific. Participation was voluntary, and individuals were paid $0.75 to complete a 5-min survey. To compensate for potential attention-check failures, we recruited 1,033 participants overall. Our sample was representative of the U.S. adult population and stratified by sex, age, and ethnicity. We excluded participants whose first language was not English (n = 54), those who failed the attention check (n = 25), and those who did not fully complete the survey (n = 38), consistent with our preregistration. This resulted in a

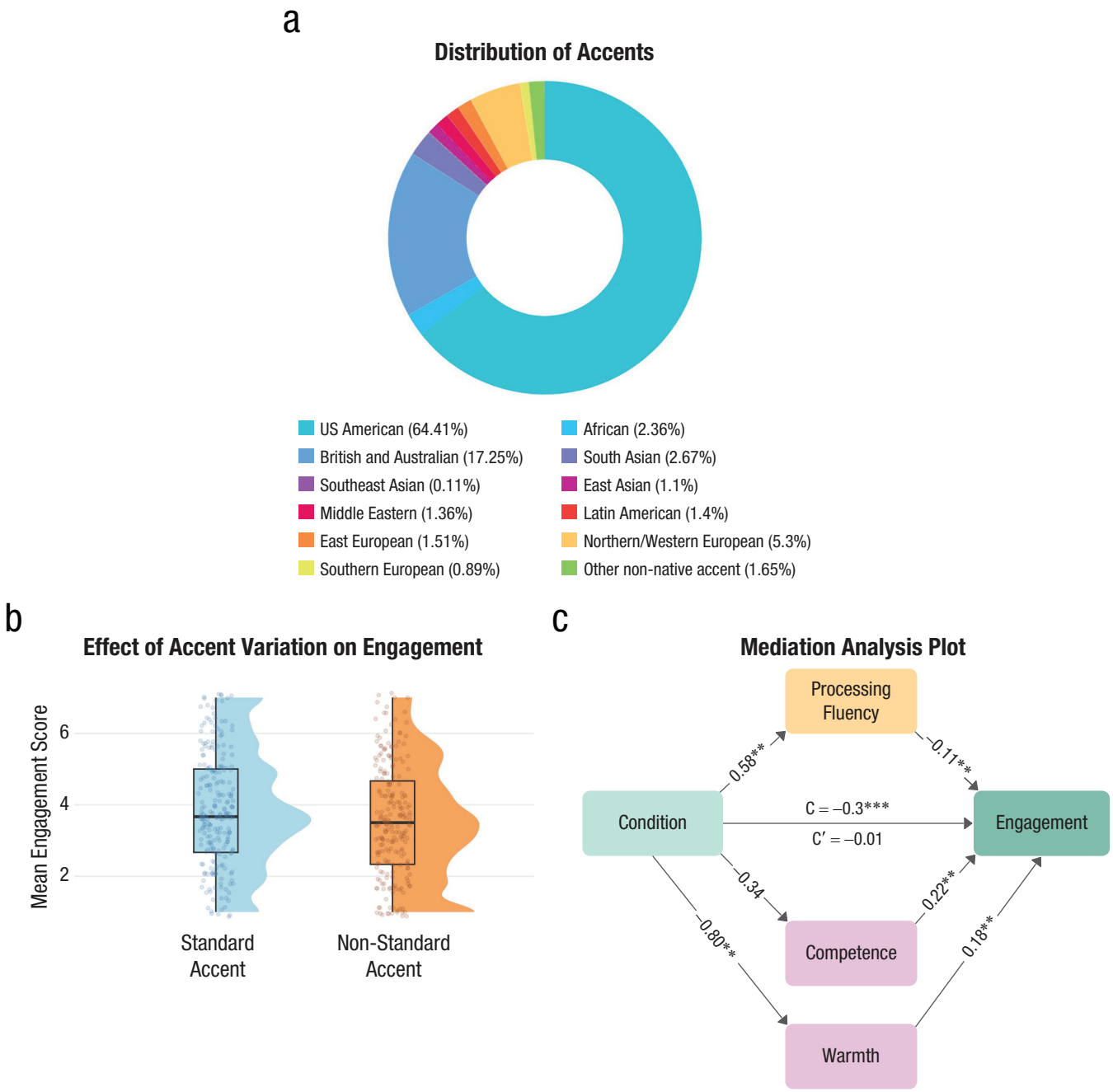

Fig. 3. Descriptive statistics, main effect, and mediation model. (a) The distribution of accents manually coded across 5,367 TED Talks for Study 1. (b) Raincloud plot describing the distribution of data points in each condition in Study 2a. (c) Mediation analysis showing that the mechanisms explaining the reduced engagement in the nonstandard accent condition are processing fluency and warmth in Study 2a. In other words, participants in the nonstandard accent condition found the speaker colder and more difficult to understand, hence decreasing engagement.

total of 916 individuals remaining for statistical analyses (M_{age} = 45.87 years, *SD* = 16.06; 458 women, 440 males, 10 nonbinary/third gender, six preferred not to say, two self-described as demiman and agender). Procedures for this study followed the same format as Study 2a: After consenting to participate and confirming that they had a functional speaker, individuals were assigned to the standard accent condition (American/Canadian accent) or nonstandard accent condition (Persian accent).

Participants listened to an audio snippet of a TED Talk in which an actor either spoke in an American/Canadian accent or a Persian accent, consistent with our

preregistration. In the same manner as Study 2a, we computed composite scores for each of our measures of interest because of the similarity of questions associated with each variable. Participants answered a series of questions related to intended engagement ($\alpha = .90$), processing fluency ($\alpha = .77$), and perceptions of the speaker's warmth ($\alpha = .93$) and competence ($\alpha = .90$). Last, individuals were asked to complete some demographic questions.

Results. The standard accent condition ($M = 2.52$, $Md = 2.00$) was rated lower than the nonstandard accent condition ($M = 6.03$, $Md = 6.00$), $t = -35.41$, $p < .001$, Hedges $g = -2.34$, 95% CI = [−2.51, −2.17], consistent with the results from Study 2a. Results of intended engagement were also similar to Study 2a in that they were significantly higher in the standard accent condition ($M = 3.87$) than in the nonstandard accent condition ($M = 3.64$), $t(907.53) = 2.17$, $p = .030$, Hedges $g = 0.14$, 95% CI = [0.01, 0.27]. The small effect sizes observed in Studies 2a and 2b were comparable to those reported in prior studies. For example, in their set of studies, Sepehri et al. (2025) used a similar data set to investigate information load on audience engagement and the mediating role of processing fluency. These authors found that as topic density increased, viewer engagement worsened ($\eta_p^2 = .11$), with processing disfluency acting as a mediator, $\eta_p^2 = .19$. Incorporating demographics such as gender, age, educational background, and political ideology into our regression framework did not eliminate the effect ($\beta = -0.22$, $SE = 0.103$, $p = .037$).

Participants' willingness to engage was lower when they found the content harder to process ($r = -.27$, $p < .001$) and stronger when they perceived the speaker as competent ($r = .66$, $p < .001$) and warm ($r = .67$, $p < .001$). In terms of mediating mechanisms, talks that were again harder to process and given by speakers perceived as cold and incompetent were least likely to receive engagement. Study 2b's mediation model was almost a spitting image of Study 2a's model, with three mediators suggesting that the entire mediation model was significant, *ab* effect through all mediators = −0.19, $SD = 0.08$, 95% CI = [−0.34, −0.04]. Additionally, warmth, bootstrapped *ab* effect = −0.09, $SD = 0.04$, 95% CI = [−0.17, −0.02], and processing fluency, bootstrapped *ab* effect = −0.05, $SD = 0.02$, 95% CI = [−0.08, −0.01], but not competence, bootstrapped *ab* effect = −0.05, $SD = 0.04$, 95% CI = [−0.14, 0.03], had significant mediating effects. The mediation model strengthened the suggestion that accent was linked to both cognitive processing and stereotyping (especially perceiving accented people as colder), which in turn were associated with lower engagement with the speaker. Notably, the mediation model was robust to controlling for the interdependence between mediators in a structural equations model (see the Supplemental Material).

General Discussion

Using a computational analysis of public speaking in the real world and a tightly controlled experiment, we provided correlational and causal evidence that speakers with a nonnative accent get less engagement. Highlighting the substantial power of accent as a conspicuous signal of cultural background and social identity (us vs. them) in the public arena, people were more likely to view and press the like button for TED Talks with native English accents (e.g., American, British) even after controlling for the demographics and fame of the speaker, topics discussed, and other non-accent-related features of their voice such as pauses.

Our social-psychological experiment simultaneously tested two mechanisms responsible for the accentism penalty effect: processing fluency and stereotyping. We found evidence for both mechanisms; however, the effect of warmth perception (friendliness, helpfulness, sincerity, and trustworthiness) was slightly stronger in our mediation model. People found it more difficult to cognitively process nonstandard accented talks, leading to reduced intention to engage with the content, consistent with the processing fluency account. Speakers with nonstandard accents were perceived as being less friendly and trustworthy, making native English-speaking perceivers less interested and less motivated to engage with them, consistent with the stereotyping account.

We highlighted our multimethodological approach to capture this effect. In our computational analyses, we combined methods from voice recognition and natural language processing, but we also relied on manual human annotations to further dissect the role of accent-based bias in online engagement. It is worth noting that our automated measure of accent is not without limitations. In our study, our primary measure of accentedness was based on phonemic and prosodic (stress, rhythm, and intonation) features. Not differentiating between prosodic and phonemic features is a limitation of the current research because prosodic features may independently drive engagement in ways that were not captured by our current approach. We recommend that future research independently measure prosodic features (e.g., pitch accenting, intonation) and phonemic features (e.g., substitution, deletion) and test their association with engagement.

Our experimental design in Study 2 complemented our computational analyses. For example, we did not know the origins of the online audiences in Study 1.

Moreover, we could not isolate the sociocognitive processes that reduced people's interest in engaging with TED Talks because we did not have direct access to viewers. In our high-powered experimental design, we recruited native English-speaking Americans residing in the United States. We further collected individual-level data about their metacognitive processing and explicit stereotyping of the speaker, enabling us to dissect the mechanisms that can explain why public speakers with nonstandard accents garner less engagement. We note, however, that our categorization of the Persian accent as not widely identifiable may have inadvertently activated stereotypes associated with non-White ethnicities, potentially influencing our results. Future replications of this work should ask participants to report the "suspected" ethnicity of the speaker in both nonstandard and standard accent conditions. Moreover, in Study 2a, processing fluency was weakly correlated with warmth ($r = -.30$) and competence ($r = -.25$), with similar results showcased in Study 2b (warmth: $r = -.23$; competence: $r = -.28$). Although the associations were weak, there was some overlap. Future studies should aim to untwine these mechanisms by testing different speakers with the same accent and varying levels of warmth and intelligibility.

Further research should examine the replicability and generalizability of our findings in other languages (e.g., public speaking in non-English languages; Blasi et al., 2022), other populations (e.g., the accentism penalty across cultural contexts; Hegarty, 2020), and other platforms (e.g., conference presentations, Talks at Google). Our studies hint at a global effect; that is, regardless of accent group, speakers with stronger accents garnered fewer views and likes (see Fig. 2). However, replicating these effects *within* countries remains an interesting area of research (Frances et al., 2018; Lev-Ari & Keysar, 2010). For example, Southern versus Californian accents in the United States have different implicit and explicit connotations; hence, perceivers' engagement may vary as a function of these stereotypes (Kinzler & DeJesus, 2013). Moreover, future research should consider the cultural dynamics of communication (e.g., conversational strategies used in specific cultures) and how they might influence engagement (Luna et al., 2008). Additionally, given topic-specific stereotypes (e.g., Indians are tech-savvy, Jewish people are good at finance), we suggest that future experimental work manipulate topic domains more intentionally in relation to group-based stereotypes to test whether and how topic-specific congruence or incongruence moderates audience judgments.

These findings have important implications for the visibility that public speakers strive for in their online engagement and dissemination of their ideas. TED Talks, for example, posts international talks online about a variety of topics (e.g., activism, art, science, technology, entrepreneurship, motivation, well-being, AI) for global distribution under the slogan "ideas worth spreading." Many of the speakers come from diverse cultural backgrounds, and English is their second (or third or fourth) language. Their nonnative accent, a verbal feature irrelevant to the "idea" discussed, can act as a hurdle to "spreading," potentially impacting their career. Our experiment can shed some light on potential solutions. One way to counteract this accent-based bias is to ensure all ideas are given a leveled ground to engage people. For example, platforms such as TED could feature talks in which the speaker has a nonnative accent. Another potential way could be to use technology to increase the ease of cognitive processing, for example, by providing subtitles in many languages, AI-assisted voice-augmentation techniques that can slightly correct mispronunciations, and natural language processing technology for real-time translation into many languages (Barrault et al., 2023). Last, educating the public about accentism as an understudied source of social bias (in addition to more known sources of bias such as racism and sexism) can be helpful. Indeed, research in social psychology has revealed that informing people about their biases can slightly change their attitudes and behaviors in a positive direction (Hahn & Gawronski, 2018).

Conclusion

Everyone has an accent; some are considered native or standard, and some are nonnative or nonstandard. In a sense, an accent is a social and cultural signal. This means it can activate preexisting stereotypes associated with social and cultural groups. In some cases, nonnative accents may also undermine effective communication. Here, we showed that public speakers, most of whom are highly competent and famous artists, scientists, physicians, and entrepreneurs from around the globe, may fail to get the attention they deserve if they speak English with a nonnative accent. In a large-scale, ecologically valid study, we used voice recognition, natural language processing, and vision models to extract audio, semantic, and visual information from more than 5,000 TED speakers. In a number of conservative tests of our prediction (i.e., adjusting for many control variables), we found that speakers with nonnative accents garnered less online engagement. To isolate and replicate this effect under experimental conditions, we complemented our work with a social-psychological experiment, confirming the main effect and showing that both stereotyping and processing fluency were associated with the engagement that accented individuals get.

Transparency

Action Editor: Antonio L. Freitas
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Author Contributions

Aliah Zewail: Conceptualization; Formal analysis; Methodology; Visualization; Writing – original draft.

Amir Sepehri: Conceptualization; Data curation; Investigation; Methodology; Project administration; Software; Writing – review & editing.

Reihane Boghrati: Conceptualization; Data curation; Formal analysis; Software; Validation; Writing – review & editing.

Mohammad Atari: Conceptualization; Methodology; Project administration; Supervision; Visualization; Writing – review & editing.

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Open Practices

Open practices for this article are described in the Research Transparency Statement section, which appears at the end of the Introduction section in the main text.

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Supplemental Material

Additional supporting information can be found at http://journals.sagepub.com/doi/suppl/10.1177/09567976251398471

Notes

1. Although previous research has used the terms "native" and "nonnative" and "standard" and "nonstandard" interchangeably to describe accents, respectively, we intentionally separated our usage. In the context of Study 1, the terms "native" and "nonnative" seemed more appropriate because we did not know what the listeners of TED Talks considered a standard accent within the English language. For Study 2, because our target population was native English-speaking Americans residing in the United States, we decided "standard" and "nonstandard" were more suitable terms.
2. We were unable to extract transcripts for 626 TED Talks.

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